

Survey Instruments In Math Anxiety Survey

1 Abbreviated Math Anxiety Scale (AMAS)

Please rate your feelings during different activities on a scale from 1 to 5. Try not to spend too much time on any one item. There are no right or wrong answers.

1 = No bad feelings, 2 = Somewhat bad, 3 = nervous, 4 = Very bad feelings, 5 = Worst feelings

1. Having to use tables in the back of a math book.
2. Thinking about a math test the day before you take it.
3. Watching the teacher work out a math problem on the board.
4. Taking a math test.
5. Being given a homework assignment of many difficult problems that is due for the next class meeting.
6. Listening to a lecture in math class. [Listening to the teacher talk for a long time in math]
7. Listening to another student explain how to do a math problem.
8. Being given a quiz on math without knowing in advance.
9. Starting a new chapter in a math book. [Starting a new topic in math]

2 Math Self-efficacy Scale

Select the response that matches how much you now agree with each statement. Try not to spend too much time on any one item. There are no right or wrong answers.

1 = Strongly disagree 2 = disagree 3 = neutral 4 = agree 5 = Strongly agree

1. I usually understand a mathematical idea quickly.
2. I have to work very hard to understand mathematics.
3. I can connect the mathematical ideas that I have learned.

3 Teaching Strategies Questionnaire

The following statements are the ways your teacher teaches mathematics. Respond to the items listed below:

5 – Always, 4 – Often, 3 – Sometimes, 2 – Seldom, 1 – Never

3.1 Cooperative-Approach

1. The teacher encourages students to work with others to generate as many alternatives as they can for the problem discussed.
2. The teacher gives students enough time to think and to investigate with others to achieve a desirable result.
3. The teacher gives students a chance to generate new concepts.
4. The teacher applies group work in the class to serve desired objectives.
5. The teacher distributes different teaching-learning tasks to students.
6. The teacher lets students have their own conversations positively.

3.2 Lecture Type

1. The teacher provides students with feedback regarding their answers at all times.
2. The teacher ends the teaching-learning situation by clarifying and discussing diagrams suitable for students.
3. The teacher trains students on generating specific answers to the questions raised to them.
4. Teachers' cognitive teaching strategies harmonize with students' learning strategies.
5. The teacher allows students to have more clarifications and explanations on a certain topic.
6. The teacher trains students to solve their problems in a comfortable way.
7. The teacher makes students take part in different roles.

3.3. Deductive Approach

1. The teacher trains students to determine the whole idea of the topic.
2. The teacher provides students with a chance to apply new knowledge in new real-life situations.

3. The teacher trains students on learning the whole concept before the specific idea.
4. Students tend to generate new information by making comparisons between their previous knowledge and the new information.
5. The teacher moves from the abstract to the concrete examples.
6. The teacher asks students to do written or verbal summaries of the information they get.
7. The teacher distributes different teaching-learning tasks to students.

3.4. Inductive Approach

1. The teacher uses specific questions to discuss the whole topic.
2. The teacher disassembles the teaching-learning material into specific tasks that need specific responses.
3. The teacher trains his students on distinguishing between different characteristics of the same concept.
4. The teacher begins by presenting the main ideas of the topic at the beginning of the class.
5. The teacher uses specific problem-solving strategies in the teaching process.
6. The teacher assigns students a specific task within a general task.
7. The teacher helps students to analyze the main idea to be used in discussing the topic as a whole.

3.5. Demonstration

1. The teacher uses direct presentation to provide students with information.
2. The teacher helps his students imitate desired models by showing them.
3. The teacher begins the teaching-learning situation by presenting a problem to students.
4. The teacher trains students to plan, observe, and evaluate their teaching activities.
5. The teacher shows students how to verify information and facts before making judgments.
6. The teacher begins with examples of the concept in the teaching-learning situation.

7. The teacher teaches students the way to identify those simple tricks to understand the lesson.

3.6. Repetitive Exercises

1. The teacher takes advantage of providing different activities to secure the teaching-learning process.
2. The teacher cares about correcting students by providing many worksheets.
3. The teacher gives similar examples during the discussion to secure the mastery of the topic.
4. The teacher takes part in training students by providing different learning activities.
5. The teacher helps students identify their own mistakes by doing similar worksheets.
6. The teacher gives students the chance to correct their mistakes by answering similar questions.
7. The teacher trains students by providing different sets of worksheets.

3.7. Integrative Approach

1. The teacher awards students for their correct answer.
2. Teacher depends on criteria in evaluating their students.
3. The teacher neglects undesired behaviors in teaching-learning situations.
4. The teacher makes use of concept maps during the teaching-learning process.
5. The teacher facilitates students to make use of the procedures that organize memory potentials (symbolizing information).
6. The teacher guides students to references such as dictionaries, encyclopedias, internet sites, etc.
7. The teacher supports students in using different learning tools for the purpose of teaching the learning process.

4 Use of technologies Scale

1: strongly disagree, 2: disagree, 3. Neutral, 4 agree, 5. Strongly agree

1. I feel apprehensive about using information technologies (ITs)
2. Technological information sounds like confusing jargon to me
3. I have avoided ITs because it is unfamiliar to me
4. I hesitate to use ITs for fear of making mistakes I cannot correct
5. ITs do not scare me at all
6. Working with ITs would make me very nervous
7. I do not feel threatened when others talk about ITs
8. I feel aggressive and hostile toward ITs
9. IT makes me feel uncomfortable
10. I get a sinking feeling when I think of trying to use ITs
11. IT makes me feel uneasy
12. IT makes me feel confused

5 Learning modalities

1. When I listen to a class lecture...
 - a. I listen very closely.
 - b. I try to be close to the speaker and watch the speaker.
 - c. I take notes during the lecture.
2. I like to solve word problems by...
 - a. talking to a friend or to myself.
 - b. using an organized approach with lists or charts.
 - c. walking, pacing, or doing something active.
3. When someone tells me numbers, but I am unable to write them down, I...
 - a. Repeat the numbers to myself out loud.

- b. visualize or see the numbers in my mind.
- c. Write the numbers in the air or on the table.
- 4. I learn something new by...
 - a. having someone explain it to me while I listen.
 - b. having someone do it for me while I watch.
 - c. doing it myself.
- 5. When I watch a movie, I remember...
 - a. everything (what was said, music, background noises).
 - b. the costumes, environment, and scenery.
 - c. how it made me feel.
- 6. When I am trying to remember something, I...
 - a. hear what was said or what sounds were around me.
 - b. visualize it happening again in my mind.
 - c. feel the way I did when it happened.
- 7. When I do not know how to spell a word, I...
 - a. sound it out.
 - b. see the word in my mind.
 - c. Write the word on paper until it looks right.
- 8. I enjoy reading when the story has...
 - a. a lot of dialogue (characters talking to each other).
 - b. a lot of descriptive words.
 - c. a lot of action.
- 9. I remember new people by...
 - a. their names.
 - b. their faces.
 - c. their actions.

10. I have a hard time concentrating when...

- a. There is a lot of noise.
- b. There are a lot of people.
- c. I am uncomfortable (too hot, too cold, uncomfortable chair, etc.).

11. When it comes to clothes, I prefer to dress...

- a. in any way, since clothes are not that important to me.
- b. well – and I have a particular style.
- c. comfortably, so I can move around easily.

12. If I cannot read aloud or get up and move around, I...

- a. Talk with a friend.
- b. look out a window.
- c. rock in my chair, tap my foot, drum my fingers, or jiggle my pencil.

6 Engagement and Resources

During the current school year at school, how often have you done the following: 1 - Very often, 2 - Often, 3 - Sometimes, 4 – Never

6.1 Engagement

- 1. Asked questions in class.
- 2. Answered questions from professors or peers in class.
- 3. Contributed to class or group discussions.
- 4. Practice exercises/examples.
- 5. Took detailed class notes.
- 6. Previewed the content before class.
- 7. Reviewed class notes and other recommended exercises
- 8. Worked with other students on the course materials.

6.2. Use of Resources

- 9. Got help from professors via email.
- 10. Went to the professor's in-person/Zoom office hours.
- 11. Used free peer tutoring through the Math Learning Center
- 12. Used private tutoring services.